OR4030 OPTIMIZATION Chapter 2
Mathematical Background for Nonlinear Optimization
2.1 Sets

Neighborhood

The *ε-neighborhood* of a point $x = (x_1, x_2, \ldots, x_n)^T \in \Re^n$ is the set of points inside the ball with center x and radius $\varepsilon > 0$:

$$N_\varepsilon(x) = \{y \in \Re^n : \|y - x\| < \varepsilon\},$$

where $\|y - x\| = \sqrt{\sum_{j=1}^{n}(y_j - x_j)^2}$ for $y = (y_1, y_2, \ldots, y_n)^T$.

Open Sets

A set S is called *open* if around every point in S there is a neighborhood that is contained in S. For example, $\Re^n$, $\emptyset$ and the open interval (a, b) in $\Re^1$ (or simply written as $\Re$) are open sets.

Closed Sets

A set S in $\Re^n$ is called *closed* if its complement S' (i.e., the set $\Re^n \setminus S$) is an open set. For example, $\Re^n$, $\emptyset$ and the closed interval $[a, b]$ in $\Re$ are closed sets.

Bounded Sets

A set is *bounded* if it can be contained in a ball of a sufficiently large radius. For example, the open interval (a, b) and the closed interval $[a, b]$ in $\Re$ are both bounded sets (here a and b are two finite numbers).

Compact Sets

A set is *compact* if it is both closed and bounded. For example, the closed interval $[a, b]$ is a compact set.

2.2 Convex Sets

A set S in $\Re^n$ is *convex* if for any elements $\bar{x}$ and $\hat{x}$ of S,

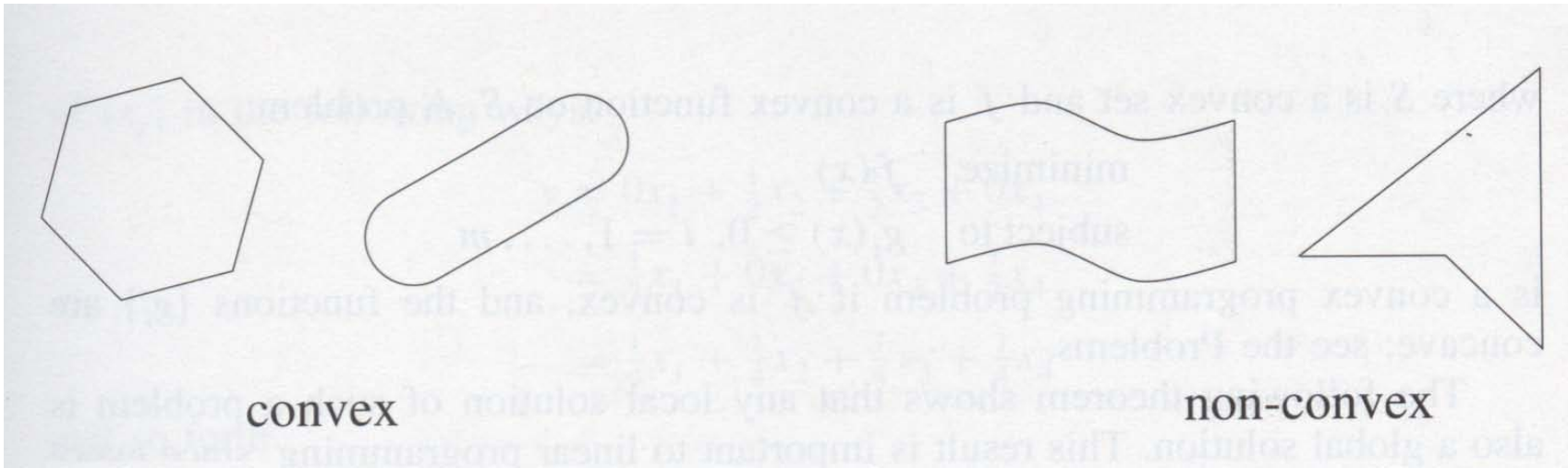
$$\lambda\bar{x} + (1-\lambda)\hat{x} \in S$$

for all values of λ such that $0 \leq \lambda \leq 1$.

The set

$$\begin{aligned} & \{\lambda\bar{x} + (1-\lambda)\hat{x} \mid 0 \leq \lambda \leq 1\} \\ = \quad & \{\hat{x} + \lambda(\bar{x} - \hat{x}) \mid 0 \leq \lambda \leq 1\} \end{aligned}$$

is the line segment joining $\bar{x}$ and $\hat{x}$. So, S is a convex set if and only if for every pair of $\bar{x}$ and $\hat{x}$ in S, the line segment joining $\bar{x}$ and $\hat{x}$ also lies in S.



Convex and Non-convex Sets

It is easy to check by the definition that the following sets are all convex sets.

- Any line:

$$L = \{x + \lambda v \mid \lambda \in \Re\}$$

for a given point (or say a vector) x and a given vector v in $\Re^n$.

- Any half line or ray:

$$L = \{x + \lambda v \mid \lambda \geq 0\}$$

for a given point x and a given vector v in $\Re^n$.

- Any closed half-space

$$F^+ = \{y \in \Re^n \mid \bar{x}^T y \geq \alpha\}$$

for a given vector $\bar{x}$ in $\Re^n$ and a given real number α.

- Any open half-space

$$F = \{y \in \Re^n \mid \bar{x}^T y > \alpha\}$$

for a given vector $\bar{x}$ in $\Re^n$ and a given real number α.

- Any closed ball

$$B^+(\bar{x}, r) = \{y \in \Re^n \mid \|y - \bar{x}\| \leq r\}$$

or open ball

$$B(\bar{x}, r) = \{y \in \Re^n \mid \|y - \bar{x}\| < r\},$$

where $\bar{x}$ is a given point in $\Re^n$, and r is a positive number.

- The intersection of a number of convex sets C_i in $\Re^n$, $i = 1, 2, \ldots, k$:

$$C = \{\bigcap_{i=1}^{k} C_i \mid C_i, i = 1, \ldots, k, \text{ are convex sets}\}.$$

- The solution set to the linear equations

$$S = \{x \in \Re^n \mid Ax = b\},$$

where $A \in \Re^{m \times n}$ (i.e., a $m \times n$ real matrix), and $b \in \Re^m$.

- For p given points $x^1, x^2, \ldots, x^p$ in $\Re^n$, any vector

$$\sum_{i=1}^{p} \lambda_i x^i = \lambda_1 x^1 + \lambda_2 x^2 + \cdots + \lambda_p x^p$$

is called a *convex combination of* $x^1, \ldots, x^p$, if all numbers λ_i satisfy that

$$\lambda_i \geq 0, \;\; i = 1, \ldots, p; \quad \text{and} \;\; \sum_{i=1}^{p} \lambda_i = 1.$$

For p given points $x^1, x^2, \ldots, x^p$ in $\Re^n$, all their convex combinations:

$$C = \{\sum_{i=1}^{p} \lambda_i x^i \mid \lambda_i \geq 0, \sum_{i=1}^{p} \lambda_i = 1\}$$

form a convex set. This set is called the convex hull of points $x^1, x^2, \ldots, x^p$.

Note that for three points x^1, x^2 and x^3 in $\Re^2$, all their convex combinations form the triangular region with x^1, x^2 and x^3 as its vertexes.

Limits

The equation

$$\lim_{x \to c} f(x) = d$$

means that as the single variable x gets very close to the number c (but is not necessary equal to c), the value of $f(x)$ gets arbitrarily close to the value d.

It is also possible that

$$\lim_{x \to c} f(x)$$

may not exist.

Continuity

A function $f(x)$ is *continuous* at a point (or say a value) c if

$$\lim_{x \to c} f(x) = f(c).$$

If $f(x)$ is not continuous at c, we say that $f(x)$ is *discontinuous* (or has a discontinuity) at c.

Differentiation

The *derivative* of $f(x)$ at a point c is defined by

$$f'(c) = \frac{df(c)}{dx} = \left.\frac{df}{dx}\right|_{x=c} = \lim_{\Delta x \to 0} \frac{f(c + \Delta x) - f(c)}{\Delta x}$$

provided that the limit exists. When the limit exists, we say that $f(x)$ is *differentiable* at c. Note also that if $f(x)$ is differentiable at c, then $f(x)$ is continuous at c.

Higher Derivatives

Furthermore, we can define

$$f^{(2)}(c) = f''(c) = \frac{d^2 f(c)}{dx^2} = \left.\frac{d^2 f}{dx^2}\right|_{x=c}$$

to be the derivative of function $f'(x)$ at point $x = c$.

Similarly, we define (if it exists) $f^{(k)}(c)$ to be the derivative of $f^{(k-1)}(x)$ at $x = c$ for $k \geq 3$.

Continuously Differentiable Functions

- $f \in C^0(S)$
  $\sim$ f is a real-valued continuous function in its domain S.

- $f \in C^1(S)$
  $\sim$ f is a real-valued continuously differentiable function in S
  $\sim$ i.e., f' exists and is continuous everywhere in S.

- $f \in C^k(S)$, where $k = 1, 2, 3, \ldots$
  $\sim$ f is a real-valued kth-order continuously differentiable function in S
  $\sim$ i.e., $f^{(k)}$ exists and is continuous everywhere in its domain

Taylor's Theorem

Suppose $f^{(k+1)}$ exists for every point on the interval $[a, b]$. Let $c \in [a, b]$. Then for every $x \in [a, b]$, there exists h between c and x with

$$f(x) = P_k(x) + R_k(x),$$

where

$$\begin{aligned} P_k(x) &= f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots \\ &\quad + \frac{f^{(k)}(c)}{k!}(x-c)^k, \\ R_k(x) &= \frac{f^{(k+1)}(h)}{(k+1)!}(x-c)^{k+1}. \end{aligned}$$

Here

$P_k(x)$ is called the *kth-order Taylor series expansion* of f about c, and

$R_k(x)$ is called the *remainder term* (or *truncation error*) associated with $P_k(x)$.

The infinite series obtained by taking the limit of $P_k(x)$ as $k \to \infty$ is called the *Taylor series* of f about c.

We can use $P_k(x)$ to approximate $f(x)$: $f(x) \approx P_k(x)$ with an error of $R_k(x)$.

2.3.2 Differential Calculus of Several Variables

Partial Derivatives and the Gradient Vector

The *partial derivative* of f with respect to the variable x_j is

$$\frac{\partial f}{\partial x_j} = \lim_{\Delta x_j \to 0} \frac{f(x_1, \ldots, x_j + \Delta x_j, \ldots, x_n) - f(x_1, \ldots, x_j, \ldots, x_n)}{\Delta x_j}.$$

The *gradient* of f is defined by:

$$\nabla f = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \vdots \\ \dfrac{\partial f}{\partial x_n} \end{bmatrix}.$$

Second-Order Partial Derivatives and the Hessian Matrix

We use the notation

$$\frac{\partial^2 f}{\partial x_i \, \partial x_j}$$

to denote a *second-order partial derivative* of f.

To find $\frac{\partial^2 f}{\partial x_i \, \partial x_j}$, we first find $\frac{\partial f}{\partial x_j}$ and then take its partial derivative with respect to x_i, i.e.,

$$\frac{\partial^2 f}{\partial x_i \, \partial x_j} = \frac{\partial}{\partial x_i} \left[\frac{\partial f}{\partial x_j} \right].$$

If the second-order partial derivatives exist and are everywhere continuous, then

$$\frac{\partial^2 f}{\partial x_i \, \partial x_j} = \frac{\partial^2 f}{\partial x_j \, \partial x_i}.$$

Suppose $f \in C^2(\Re^n)$. The *Hessian* matrix of f is defined by

$$\nabla^2 f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial {x_1}^2} & \cdots & \dfrac{\partial^2 f}{\partial x_n \, \partial x_1} \\ \vdots & & \vdots \\ \dfrac{\partial^2 f}{\partial x_1 \, \partial x_n} & \cdots & \dfrac{\partial^2 f}{\partial {x_n}^2} \end{bmatrix}.$$

Note that the Hessian matrix of f is symmetric if all second order partial derivatives are continuous.

Taylor's Theorem

Suppose that $f \in C^3(S)$, where

$$S = \{x : a_j \leq x_j \leq b_j, j = 1, 2, \ldots, n\}.$$

Let $c = (c_1, c_2, \ldots, c_n)^T \in S$. For every $x \in S$,

$$f(x) \approx f(c) + \nabla f(c)^T (x - c) + \frac{1}{2}(x - c)^T \nabla^2 f(c)(x - c).$$

If we want to change the above approximate equality to an exact one, then

$$f(x) = f(c) + \nabla f(c)^T (x - c) + \frac{1}{2}(x - c)^T \nabla^2 f(\xi)(x - c),$$

where ξ is a point in the line segment linking x and c, but its exact location is unknown.

Example

For the function

$$f(x) = x_1^3 + 5x_1^2x_2 + 7x_1x_2^2 + 2x_2^3$$

where $x = (x_1, x_2)^T$, consider its approximate function values near the point $c = (-2, 3)^T$.

The gradient of this function is

$$\nabla f(x) = \begin{bmatrix} 3x_1^2 + 10x_1x_2 + 7x_2^2 \\ 5x_1^2 + 14x_1x_2 + 6x_2^2 \end{bmatrix}$$

and the Hessian matrix is

$$\nabla^2 f(x) = \begin{bmatrix} 6x_1 + 10x_2 & 10x_1 + 14x_2 \\ 10x_1 + 14x_2 & 14x_1 + 12x_2 \end{bmatrix}.$$

At the point c, $f, \nabla f$ and $\nabla^2 f$ become

$$f(c) = -20, \quad \nabla f(c) = \begin{bmatrix} 15 \\ -10 \end{bmatrix} \quad \text{and} \quad \nabla^2 f(c) = \begin{bmatrix} 18 & 22 \\ 22 & 8 \end{bmatrix}.$$

By Taylor's Theorem,

$$\begin{aligned} f(x) \quad &\approx \quad -20 + \begin{bmatrix} 15 & -10 \end{bmatrix} \begin{bmatrix} x_1 + 2 \\ x_2 - 3 \end{bmatrix} \\ &\qquad + \frac{1}{2} \begin{bmatrix} x_1 + 2 & x_2 - 3 \end{bmatrix} \begin{bmatrix} 18 & 22 \\ 22 & 8 \end{bmatrix} \begin{bmatrix} x_1 + 2 \\ x_2 - 3 \end{bmatrix} \end{aligned}$$

Thus, for example,

$$f(-1.9, 3.2) \approx -19.81.$$

The true value is $f(-1.9, 3.2) = -19.755$. So the approximation is accurate to three digits.